

Permutations

Date _____ Period _____

List all possible permutations.

1) ☺, ☀, ♥

2) ☺, ☀

3) 1, 2

4) A, B, C, taken two at a time

Evaluate each expression.5) ${}_8P_4$ 6) ${}_8P_5$ 7) ${}_7P_5$ 8) ${}_{10}P_3$ 9) ${}_{10}P_5 - 5$ 10) ${}_{10}P_3$ **Find the number of unique permutations of the letters in each word.**

11) EVERY

12) SOLO

13) BANDING

14) STRUTS

15) THERMOMETER

16) BILLIONAIRE

Critical thinking questions:

17) Name a season of the year, where if you rearrange the letters there are 360 unique permutations.

18) Simplify ${}_xP_{x-2}$